

$$\Rightarrow 3x^2 + (6 - 3x)^2 = 36 \Rightarrow 12x^2 - 36x = 0 \Rightarrow x = 0, 3$$

Putting $x = 0$ and 3 respectively in $y = \frac{6 - 3x}{\sqrt{3}}$, we get $y = 2\sqrt{3}$, and $y = -\sqrt{3}$ respectively.

Hence, the coordinates of the third vertex B are $(0, 2\sqrt{3})$ or $(3, -\sqrt{3})$.

EXAMPLE 4 If the segments joining the points $A(a, b)$ and $B(c, d)$ subtends an angle θ at the origin,

prove that: $\cos \theta = \frac{ac + bd}{\sqrt{(a^2 + b^2)(c^2 + d^2)}}$.

SOLUTION Let O be the origin. Then,

$$OA^2 = a^2 + b^2, OB^2 = c^2 + d^2 \text{ and, } AB^2 = (c - a)^2 + (d - b)^2$$

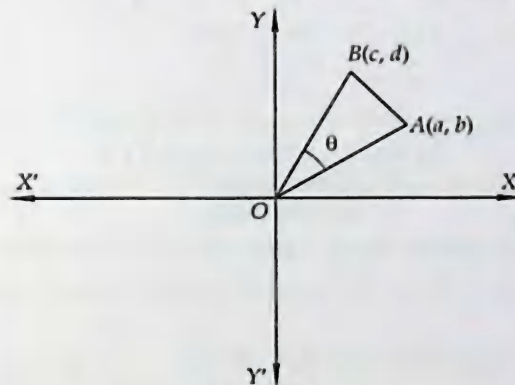


Fig. 22.6

Using cosine formula in $\triangle OAB$, we have

$$AB^2 = OA^2 + OB^2 - 2(OA)(OB) \cos \theta$$

$$\Rightarrow (c - a)^2 + (d - b)^2 = a^2 + b^2 + c^2 + d^2 - 2\sqrt{a^2 + b^2}\sqrt{c^2 + d^2} \cos \theta$$

$$\Rightarrow c^2 + a^2 - 2ac + d^2 + b^2 - 2bd = a^2 + b^2 + c^2 + d^2 - 2\sqrt{a^2 + b^2}\sqrt{c^2 + d^2} \cos \theta$$

$$\Rightarrow 2(ac + bd) = 2\sqrt{a^2 + b^2}\sqrt{c^2 + d^2} \cos \theta$$

$$\Rightarrow \cos \theta = \frac{ac + bd}{\sqrt{a^2 + b^2}\sqrt{c^2 + d^2}}$$

EXAMPLE 5 Find the coordinates of the circumcentre of the triangle whose vertices are $(8, 6)$, $(8, -2)$ and $(2, -2)$. Also, find its circum-radius.

SOLUTION Let $A(8, 6)$, $B(8, -2)$ and $C(2, -2)$ be the vertices of the given triangle and let $P(x, y)$ be the circumcentre of this triangle. Then, $PA^2 = PB^2 = PC^2$

$$\text{Now, } PA^2 = PB^2$$

$$\Rightarrow (x - 8)^2 + (y - 6)^2 = (x - 8)^2 + (y + 2)^2$$

$$\Rightarrow x^2 + y^2 - 16x - 12y + 100 = x^2 + y^2 - 16x + 4y + 68$$

$$\Rightarrow 16y = 32 \Rightarrow y = 2$$

$$\text{Again, } PB^2 = PC^2$$

$$\Rightarrow (x - 8)^2 + (y + 2)^2 = (x - 2)^2 + (y + 2)^2$$

$$\Rightarrow x^2 + y^2 - 16x + 4y + 68 = x^2 + y^2 - 4x + 4y + 8$$

$$\Rightarrow 12x = 60 \Rightarrow x = 5.$$

So, the coordinates of the circumcentre P are $(5, 2)$.

Also, Circum-radius = $PA = PB = PC = \sqrt{(5-8)^2 + (2-6)^2} = 5$.

EXAMPLE 6 The vertices of a triangle are $A(1, 1)$, $B(4, 5)$ and $C(6, 13)$. Find $\cos A$.

SOLUTION We know that: $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, where $a = BC$, $b = CA$ and $c = AB$ are the sides of the triangle ABC .

Here,

$$a = BC = \sqrt{(4-6)^2 + (5-13)^2} = \sqrt{68}, \quad b = CA = \sqrt{(6-1)^2 + (13-1)^2} = \sqrt{169} = 13,$$

$$\text{and, } c = AB = \sqrt{(4-1)^2 + (5-1)^2} = 5$$

$$\therefore \cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{169 + 25 - 68}{2 \times 13 \times 5} = \frac{63}{65}$$

EXAMPLE 7 Let the opposite angular points of a square be $(3, 4)$ and $(1, -1)$. Find the coordinates of the remaining angular points.

SOLUTION Let $ABCD$ be a square and let $A(3, 4)$ and $C(1, -1)$ be the given angular points. Let $B(x, y)$ be the unknown vertex.

Then, $AB = BC$

$$\Rightarrow AB^2 = BC^2$$

$$\Rightarrow (x-3)^2 + (y-4)^2 = (x-1)^2 + (y+1)^2$$

$$\Rightarrow 4x + 10y - 23 = 0$$

$$\Rightarrow x = \frac{23-10y}{4} \quad \dots(i)$$

Applying Pythagoras Theorem in triangle ABC , we obtain

$$AB^2 + BC^2 = AC^2$$

$$\Rightarrow (x-3)^2 + (y-4)^2 + (x-1)^2 + (y+1)^2 = (3-1)^2 + (4+1)^2$$

$$\Rightarrow x^2 + y^2 - 4x - 3y - 1 = 0 \quad \dots(ii)$$

Substituting the value of x from (i) into (ii), we get

$$\left(\frac{23-10y}{4}\right)^2 + y^2 - (23-10y) - 3y - 1 = 0$$

$$\Rightarrow 4y^2 - 12y + 5 = 0 \Rightarrow (2y-1)(2y-5) = 0 \Rightarrow y = \frac{1}{2} \text{ or } \frac{5}{2}$$

Putting $y = \frac{1}{2}$ and $y = \frac{5}{2}$ respectively in (i), we get $x = \frac{9}{2}$ and $x = -\frac{1}{2}$ respectively.

Hence, the required vertices of the square are $(9/2, 1/2)$ and $(-1/2, 5/2)$.

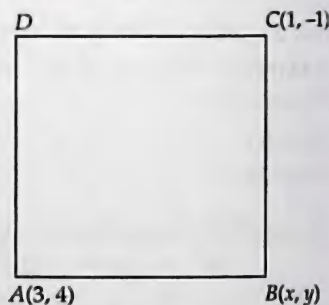


Fig. 22.7

22.3 AREA OF A TRIANGLE

THEOREM The area of a triangle, the coordinates of whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is the absolute value of

$$\frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right] \text{ or, } \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

NOTE 1 Sign of area: If the points A, B, C are plotted in the two dimensional plane and three points are taken in the anticlockwise sense then the area calculated of the triangle ABC will be positive while if the points are taken in clockwise sense then the area calculated will be negative. But, if the points are taken arbitrarily, then the area calculated may be positive or negative, the numerical value being the same in both cases. In case the area calculated is negative we will consider it positive.

NOTE 2 To find the area of a polygon we divide it into triangles and take numerical value of the area of each of the triangles.

CONDITION OF COLLINEARITY OF THREE POINTS Three points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are collinear iff

(i) Area of $\Delta ABC = 0$

$$\text{i.e. } \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0 \quad \text{or,} \quad \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

OR

(ii) $AB + BC = AC$ or, $AC + BC = AB$ or, $AC + AB = BC$

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I ON FINDING THE AREA OF A TRIANGLE WHEN COORDINATES OF ITS VERTICES ARE GIVEN

EXAMPLE 1 Prove that the area of the triangle whose vertices are $(t, t-2)$, $(t+2, t+2)$ and $(t+3, t)$ is independent of t .

SOLUTION Let $A = (x_1, y_1) = (t, t-2)$, $B = (x_2, y_2) = (t+2, t+2)$ and $C = (x_3, y_3) = (t+3, t)$ be the vertices of the given triangle. Then,

$$\begin{aligned} \text{Area of } \Delta ABC &= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| \\ &= \frac{1}{2} |t(t+2-t) + (t+2)(t-t+2) + (t+3)(t-2-t-2)| \\ &= \frac{1}{2} |2t + 2t + 4 - 4t - 12| = |-4| = 4 \text{ sq. units.} \end{aligned}$$

Clearly, area of ΔABC is independent of t .

Type II ON FINDING THE AREA OF A QUADRILATERAL WHEN COORDINATES OF ITS VERTICES ARE GIVEN

EXAMPLE 2 Find the area of the quadrilateral $ABCD$ whose vertices are respectively $A(1, 1)$, $B(7, -3)$, $C(12, 2)$ and $D(7, 21)$.

SOLUTION Clearly, Area of quadrilateral $ABCD = |\text{Area of } \Delta ABC| + |\text{Area of } \Delta ACD|$
Now,

$$\begin{aligned} \text{Area of } \Delta ABC &= \frac{1}{2} |1 \times (-3-2) + 7 \times (2-1) + 12 \times (1+3)| = \frac{1}{2} |(-5+7+48)| \\ &= 25 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of } \Delta ACD &= \frac{1}{2} |1 \times (2-21) + 12 \times (21-1) + 7 \times (1-2)| = \frac{1}{2} |(-19+240-7)| \\ &= 107 \text{ sq. units} \end{aligned}$$

$\therefore$ Area of quadrilateral $ABCD = 25 + 107 = 132 \text{ sq. units.}$

Type III ON COLLINEARITY OF THREE POINTS

EXAMPLE 3 Prove that the points $(a, b+c)$, $(b, c+a)$ and $(c, a+b)$ are collinear.

SOLUTION Let $A = (x_1, y_1) = (a, b+c)$, $B = (x_2, y_2) = (b, c+a)$ and $C = (x_3, y_3) = (c, a+b)$ be three given points. Then,

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)$$

$$\begin{aligned}
 &= a \{(c+a) - (a+b)\} + b \{(a+b) - (b+c)\} + c \{(b+c) - (c+a)\} \\
 &= a(c-b) + b(a-c) + c(b-a) = 0
 \end{aligned}$$

Hence, the given points are collinear.

Type IV ON FINDING THE DESIRED RESULT OR UNKNOWN WHEN THREE POINTS ARE COLLINEAR

EXAMPLE 4 For what value of k are the points $(k, 2-2k)$, $(-k+1, 2k)$ and $(-4-k, 6-2k)$ are collinear?

SOLUTION Let three given points be $A = (x_1, y_1) = (k, 2-2k)$, $B = (x_2, y_2) = (-k+1, 2k)$ and $C = (x_3, y_3) = (-4-k, 6-2k)$.

If the given points are collinear, then

$$\begin{aligned}
 &x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0 \\
 \Rightarrow &k(2k - 6 + 2k) + (-k+1)(6 - 2k - 2 + 2k) + (-4-k)(2 - 2k - 2k) = 0 \\
 \Rightarrow &k(4k - 6) - 4(k-1) + (4+k)(4k-2) = 0 \\
 \Rightarrow &4k^2 - 6k - 4k + 4 + 4k^2 + 14k - 8 = 0 \\
 \Rightarrow &8k^2 + 4k - 4 = 0 \Rightarrow 2k^2 + k - 1 = 0 \Rightarrow (2k-1)(k+1) = 0 \Rightarrow k = 1/2 \text{ or } k = -1.
 \end{aligned}$$

Hence, the given points are collinear for $k = 1/2$ or $k = -1$.

LEVEL-2

Type V MIXED PROBLEMS BASED UPON THE CONCEPT OF AREA OF A TRIANGLE

EXAMPLE 5 If the vertices of a triangle have integral coordinates, prove that the triangle cannot be equilateral.

SOLUTION Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a triangle ABC , where $x_i, y_i, i = 1, 2, 3$ are integers. Then, the area of ΔABC is given by

$$\Delta = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$\Rightarrow \Delta = \text{A rational number} \quad [\because x_i, y_i \text{ are integers}]$$

If possible, let the triangle ABC be an equilateral triangle, then its area is given by

$$\Delta = \frac{\sqrt{3}}{4} (\text{Side})^2 = \frac{\sqrt{3}}{4} (AB)^2 \quad [\because AB = BC = CA]$$

$$\Rightarrow \Delta = \frac{\sqrt{3}}{4} (\text{A positive integer}) \quad [\because \text{Vertices are integers} \therefore AB^2 \text{ is a positive integer}]$$

$$\Rightarrow \Delta = \text{An irrational number}$$

This is a contradiction to the fact that the area is a rational number. Hence, the triangle cannot be equilateral.

EXAMPLE 6 If the coordinates of two points A and B are $(3, 4)$ and $(5, -2)$ respectively. Find the coordinates of any point P , if $PA = PB$ and Area of $\Delta PAB = 10$.

SOLUTION Let the coordinates of P be (x, y) . Then,

$$PA = PB$$

$$\Rightarrow PA^2 = PB^2$$

$$\Rightarrow (x-3)^2 + (y-4)^2 = (x-5)^2 + (y+2)^2 \Rightarrow x-3y-1=0 \quad \dots(i)$$

Now, Area of $\Delta PAB = 10$

$$\Rightarrow \frac{1}{2} \begin{vmatrix} x & y & 1 \\ 3 & 4 & 1 \\ 5 & -2 & 1 \end{vmatrix} = \pm 10$$

$$\Rightarrow 6x + 2y - 26 = \pm 20$$

$$\Rightarrow 6x + 2y - 46 = 0 \text{ or, } 6x + 2y - 6 = 0$$

$$\Rightarrow 3x + y - 23 = 0 \text{ or, } 3x + y - 3 = 0$$

...(ii)

Solving $x - 3y - 1 = 0$ and $3x + y - 23 = 0$, we get: $x = 7, y = 2$.

Solving $x - 3y - 1 = 0$ and $3x + y - 3 = 0$, we get: $x = 1, y = -0$.

Thus, the coordinates of P are $(7, 2)$ or $(1, 0)$.

EXAMPLE 7 The coordinates of A, B, C are $(6, 3), (-3, 5)$ and $(4, -2)$ respectively and P is any point (x, y) . Show that the ratio of the areas of triangles PBC and ABC is $\left| \frac{x + y - 2}{7} \right|$.

SOLUTION We have,

$$\begin{aligned} \frac{\text{Area of } \triangle PBC}{\text{Area of } \triangle ABC} &= \frac{\frac{1}{2} |x(5 + 2) + (-3)(-2 - y) + 4(y - 5)|}{\frac{1}{2} |6(5 + 2) + (-3)(-2 - 3) + 4(3 - 5)|} \\ \Rightarrow \frac{\text{Area of } \triangle PBC}{\text{Area of } \triangle ABC} &= \frac{|7x + 7y - 14|}{|42 + 15 - 8|} = \frac{7|x + y - 2|}{49} = \left| \frac{x + y - 2}{7} \right| \end{aligned}$$

22.4 SECTION FORMULAE

- (i) The coordinates of the point P which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m : n$ are

$$\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

NOTE 1 If P is the mid-point of AB , then it divides AB in the ratio $1 : 1$, so its coordinates

$$\text{are } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

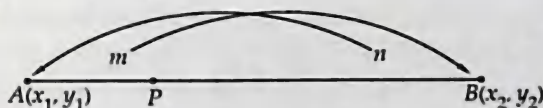


Fig. 22.8

NOTE 2 The following diagram will help to remember the section formula.

- (ii) The coordinates of the point which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ externally in the ratio $m : n$ are

$$\left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n} \right)$$

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I ON FINDING THE SECTION POINT WHEN THE SECTION RATIO IS GIVEN

EXAMPLE 1 Find the coordinates of points on the line joining the point $P(3, -4)$ and $Q(-2, 5)$ that is twice as far from P as from Q .

SOLUTION Let $R(x, y)$ be the required point. Then, $PR = 2 \cdot RQ$ (given) i.e. $PR : RQ = 2 : 1$. Thus, R divides PQ internally or externally in the ratio $2 : 1$.

If R divides PQ internally in the ratio $2 : 1$, then the coordinates of R are given by

$$x = \frac{2 \times -2 + 1 \times 3}{2 + 1} \text{ and } y = \frac{2 \times 5 + 1 \times -4}{2 + 1} \Rightarrow x = -\frac{1}{3} \text{ and } y = 2.$$

So, the coordinates of R are $(-1/3, 2)$

If R divides PQ externally in the ratio $2 : 1$, then the coordinates of R are given by

$$x = \frac{2 \times -2 - 1 \times 3}{2 - 1} \text{ and } y = \frac{2 \times 5 - 1 \times -4}{2 - 1} \Rightarrow x = -7 \text{ and } y = 14.$$

Hence, the coordinates of R are $(-7, 14)$.

Type II ON FINDING THE SECTION RATIO OR AN END POINT OF THE SEGMENT WHEN THE SECTION POINT IS GIVEN

EXAMPLE 2 Determine the ratio in which the line $3x + y - 9 = 0$ divides the segment joining the points $(1, 3)$ and $(2, 7)$.

SOLUTION Suppose the line $3x + y - 9 = 0$ divides the line segment joining $A(1, 3)$ and $B(2, 7)$ in the ratio $k : 1$ at point C . Then, the coordinates of C are $\left(\frac{2k+1}{k+1}, \frac{7k+3}{k+1}\right)$. But, C lies on the line $3x + y - 9 = 0$.

$$\therefore 3\left(\frac{2k+1}{k+1}\right) + \frac{7k+3}{k+1} - 9 = 0 \Rightarrow 6k + 3 + 7k + 3 - 9k - 9 = 0 \Rightarrow k = \frac{3}{4}$$

So, the required ratio is $3 : 4$ internally.

Type III ON DETERMINATION OF THE TYPE OF A GIVEN QUADRILATERAL

EXAMPLE 3 Prove that: $(4, -1)$, $(6, 0)$, $(7, 2)$ and $(5, 1)$ are the vertices of a rhombus. Is it a square?

SOLUTION Let the given points be A, B, C and D respectively. Then,

$$\text{Coordinates of the mid-point of } AC \text{ are } \left(\frac{4+7}{2}, \frac{-1+2}{2}\right) = \left(\frac{11}{2}, \frac{1}{2}\right)$$

$$\text{Coordinates of the mid-point of } BD \text{ are } \left(\frac{6+5}{2}, \frac{0+1}{2}\right) = \left(\frac{11}{2}, \frac{1}{2}\right)$$

Thus, AC and BD have the same mid-point. Hence, $ABCD$ is a parallelogram.

$$\text{Now, } AB = \sqrt{(6-4)^2 + (0+1)^2} = \sqrt{5}, BC = \sqrt{(7-6)^2 + (2-0)^2} = \sqrt{5}$$

$$\therefore AB = BC$$

So, $ABCD$ is a parallelogram whose adjacent sides are equal. Hence, $ABCD$ is a rhombus.

$$\text{Now, } AC = \sqrt{(7-4)^2 + (2+1)^2} = 3\sqrt{2}, \text{ and } BD = \sqrt{(6-5)^2 + (0-1)^2} = \sqrt{2}.$$

Clearly, $AC \neq BD$. So, $ABCD$ is not a square.

Type IV ON FINDING THE UNKNOWN VERTEX FROM GIVEN POINTS

EXAMPLE 4 If the coordinates of the mid-points of the sides of a triangle are $(1, 2)$, $(0, -1)$ and $(2, -1)$. Find the coordinates of its vertices.

SOLUTION Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of $\triangle ABC$. Let $D(1, 2)$, $E(0, -1)$, and $F(2, -1)$ be the mid-points of sides BC , CA and AB respectively.

Since D is the mid-point of BC .

$$\therefore \frac{x_2 + x_3}{2} = 1 \text{ and } \frac{y_2 + y_3}{2} = 2 \Rightarrow x_2 + x_3 = 2 \text{ and } y_2 + y_3 = 4 \quad \dots(i)$$

Similarly, E and F are the mid-points of CA and AB respectively.

$$\therefore \frac{x_1 + x_3}{2} = 0 \text{ and } \frac{y_1 + y_3}{2} = -1 \Rightarrow x_1 + x_3 = 0 \text{ and } y_1 + y_3 = -2 \quad \dots(ii)$$

$$\text{and, } \frac{x_1 + x_2}{2} = 2 \text{ and } \frac{y_1 + y_2}{2} = -1 \Rightarrow x_1 + x_2 = 4 \text{ and } y_1 + y_2 = -2 \quad \dots(iii)$$

From (i), (ii) and (iii), we get

$$(x_2 + x_3) + (x_1 + x_3) + (x_1 + x_2) = 2 + 0 + 4 \text{ and } (y_2 + y_3) + (y_1 + y_3) + (y_1 + y_2) = 4 - 2 - 2$$

$$\Rightarrow x_1 + x_2 + x_3 = 3 \text{ and } y_1 + y_2 + y_3 = 0 \quad \dots(\text{iv})$$

From (i) and (iv), we get

$$x_1 + 2 = 3 \text{ and } y_1 + 4 = 0 \Rightarrow x_1 = 1 \text{ and } y_1 = -4$$

So, the coordinates of A are (1, -4).

From (ii) and (iv), we get

$$x_2 + 0 = 3 \text{ and } y_2 - 2 = 0 \Rightarrow x_2 = 3 \text{ and } y_2 = 2$$

So, coordinates of B are (3, 2).

From (iii) and (iv), we get

$$x_3 + 4 = 3 \text{ and } y_3 - 2 = 0 \Rightarrow x_3 = -1 \text{ and } y_3 = 2$$

So, coordinates of C are (-1, 2).

Hence, the vertices of the triangle ABC are A (1, -4), B (3, 2) and C (-1, 2).

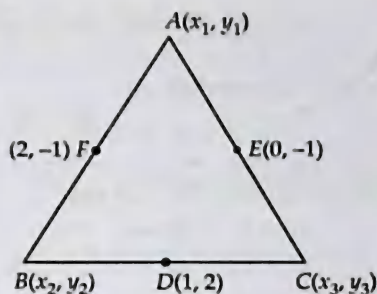


Fig. 22.9

22.5 CENTROID, IN-CENTRE AND EX-CENTRES OF A TRIANGLE

(i) The coordinates of the centroid of the triangle whose vertices are (x_1, y_1) and (x_2, y_2) are (x_3, y_3) are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$.

(ii) The coordinates of the in-centre of a triangle whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are $\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$, where $a = BC$, $b = CA$ and $c = AB$.

(iii) Let $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ be the vertices of the triangle ABC, and let a, b, c be the lengths of the sides BC, CA, AB respectively. The circle which touches the side BC and two sides AB and AC produced is called the escribed circle opposite to the angle A. The bisectors of the external angles B and C meet at a point I_1 which is the centre of the escribed circle opposite to the angle A. The coordinates of I_1 are given by

$$\left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c} \right)$$

The coordinates of I_2 and I_3 (centres of escribed circles opposite to the angles B and C respectively) are given by

$$I_2 \left(\frac{ax_1 - bx_2 + cx_3}{a - b + c}, \frac{ay_1 - by_2 + cy_3}{a - b + c} \right) \text{ and } I_3 \left(\frac{ax_1 + bx_2 - cx_3}{a + b - c}, \frac{ay_1 + by_2 - cy_3}{a + b - c} \right)$$

respectively.

LEVEL-1

EXAMPLE 1 If the coordinates of the mid-points of the sides of a triangle are (1, 1), (2, -3) and (3, 4). Find its (i) centroid (ii) in-centre.

SOLUTION Let $P(1, 1)$, $Q(2, -3)$, $R(3, 4)$ be the mid-points of sides AB, BC and CA respectively of triangle ABC. Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of triangle ABC. Then, P is the mid-point of AB.

$$\therefore \frac{x_1 + x_2}{2} = 1, \frac{y_1 + y_2}{2} = 1 \Rightarrow x_1 + x_2 = 2 \text{ and } y_1 + y_2 = 2 \quad \dots(\text{i})$$

Q is the mid-point of BC.

$$\therefore \frac{x_2 + x_3}{2} = 2, \frac{y_2 + y_3}{2} = -3 \Rightarrow x_2 + x_3 = 4 \text{ and } y_2 + y_3 = -6 \quad \dots(ii)$$

R is the mid-point of AC

$$\therefore \frac{x_1 + x_3}{2} = 3 \text{ and } \frac{y_1 + y_3}{2} = 4 \Rightarrow x_1 + x_3 = 6 \text{ and } y_1 + y_3 = 8 \quad \dots(iii)$$

From (i), (ii) and (iii), we get

$$\begin{aligned} x_1 + x_2 + x_2 + x_3 + x_1 + x_3 &= 2 + 4 + 6 \text{ and } y_1 + y_2 + y_2 + y_3 + y_1 + y_3 = 2 - 6 + 8 \\ \Rightarrow x_1 + x_2 + x_3 &= 6 \text{ and } y_1 + y_2 + y_3 = 2 \quad \dots(iv) \end{aligned}$$

From (i) and (iv), we get

$$x_3 + 2 = 6 \text{ and } 2 + y_3 = 2 \Rightarrow x_3 = 4, y_3 = 0$$

So, the coordinates of C are (4, 0).

From (ii) and (iv), we get

$$x_1 + 4 = 6 \text{ and } y_1 - 6 = 2 \Rightarrow x_1 = 2, y_1 = 8$$

So, coordinates of A are (2, 8).

From (iii) and (iv), we get

$$x_2 + 6 = 6 \text{ and } y_2 + 8 = 2 \Rightarrow x_2 = 0 \text{ and } y_2 = -6$$

So, coordinates of B are (0, -6).

$$\text{Now, } a = BC = \sqrt{(4-0)^2 + (0+6)^2} = \sqrt{52} = 2\sqrt{13}$$

$$b = AC = \sqrt{(4-2)^2 + (0-8)^2} = \sqrt{68} = 2\sqrt{17}$$

$$\text{and, } c = AB = \sqrt{(2-0)^2 + (8+6)^2} = \sqrt{200} = 10\sqrt{2}$$

The coordinates of the in-centre of the triangle ABC are

$$\begin{aligned} &\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right) \\ \text{or, } &\left(\frac{(2\sqrt{13})2 + (2\sqrt{17})0 + (10\sqrt{2})4}{2\sqrt{13} + 2\sqrt{17} + 10\sqrt{2}}, \frac{(2\sqrt{13})8 + (2\sqrt{17})(-6) + (10\sqrt{2})0}{2\sqrt{13} + 2\sqrt{17} + 10\sqrt{2}} \right) \\ \text{or, } &\left(\frac{4\sqrt{13} + 40\sqrt{2}}{2\sqrt{13} + 2\sqrt{17} + 10\sqrt{2}}, \frac{16\sqrt{13} - 12\sqrt{17}}{2\sqrt{13} + 2\sqrt{17} + 10\sqrt{2}} \right) \\ \text{or, } &\left(\frac{2\sqrt{13} + 20\sqrt{2}}{\sqrt{13} + \sqrt{17} + 5\sqrt{2}}, \frac{8\sqrt{13} - 6\sqrt{17}}{\sqrt{13} + \sqrt{17} + 5\sqrt{2}} \right) \end{aligned}$$

EXAMPLE 2 Two vertices of a triangle are (3, -5) and (-7, 4). If its centroid is (2, -1), find the third vertex.

SOLUTION Let the coordinates of the third vertex be (x, y). Then,

$$\frac{x+3-7}{3} = 2 \text{ and } \frac{y-5+4}{3} = -1$$

$$\Rightarrow x-4=6 \text{ and } y-1=-3 \Rightarrow x=10 \text{ and } y=-2$$

Thus, the coordinates of the third vertex are (10, -2).

EXERCISE 22.1

LEVEL-1

1. If the line segment joining the points P (x_1, y_1) and Q (x_2, y_2) subtends an angle α at the origin O, prove that: $OP \cdot OQ \cos \alpha = x_1 x_2 + y_1 y_2$.

- The vertices of a triangle ABC are $A(0, 0)$, $B(2, -1)$ and $C(9, 2)$. Find $\cos B$.
- Four points $A(6, 3)$, $B(-3, 5)$, $C(4, -2)$ and $D(x, 3x)$ are given in such a way that $\frac{\Delta DBC}{\Delta ABC} = \frac{1}{2}$, find x .
- The points $A(2, 0)$, $B(9, 1)$, $C(11, 6)$ and $D(4, 4)$ are the vertices of a quadrilateral $ABCD$. Determine whether $ABCD$ is a rhombus or not.
- Find the coordinates of the centre of the circle inscribed in a triangle whose vertices are $(-36, 7)$, $(20, 7)$ and $(0, -8)$.
- The base of an equilateral triangle with side $2a$ lies along the y -axis such that the mid-point of the base is at the origin. Find the vertices of the triangle. [NCERT]
- Find the distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ when (i) PQ is parallel to the y -axis
(ii) PQ is parallel to the x -axis. [NCERT]
- Find a point on the x -axis, which is equidistant from the point $(7, 6)$ and $(3, 4)$. [NCERT]

ANSWERS

- $\frac{-11}{\sqrt{290}}$
- $\frac{11}{8}$
- No
- $(-1, 0)$
- $(0, a)$, $(0, -a)$ and $(-\sqrt{3}a, 0)$ or $(0, a)$, $(0, -a)$ and $(\sqrt{3}a, 0)$
- (i) $|y_2 - y_1|$ (ii) $|x_2 - x_1|$
- $(15/2, 0)$

HINTS TO NCERT & SELECTED PROBLEM

- Let BC be the base of equilateral triangle ABC . It is given that $BC = 2a$ and mid-point of BC is at the origin. Therefore, coordinates of B and C are $(0, a)$ and $(0, -a)$ respectively. As the triangle ABC is equilateral. So, vertex A lies on x -axis. Let its coordinates be $(x, 0)$.

Also, $AB = BC = AC \Rightarrow AB = AC = 2a$

Using Pythagoras theorem in ΔAOB , we get

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow (2a)^2 = OA^2 + a^2$$

$$\Rightarrow OA = \sqrt{3}a \text{ So, coordinates of } A \text{ are } (-\sqrt{3}a, 0). \text{ Similarly, coordinates of } A' \text{ are } (\sqrt{3}a, 0).$$

Hence, the coordinates of the vertices of triangle are $A(\sqrt{3}a, 0)$, $B(0, a)$ and $C(0, -a)$ or, $A'(-\sqrt{3}a, 0)$, $B(0, a)$ and $C(0, -a)$.

- (i) If PQ is parallel to y -axis, then $x_1 = x_2$
and, $PQ = \text{Absolute value of the difference of } y\text{-coordinates of } P \text{ and } Q = |y_2 - y_1|$.
(ii) If PQ is parallel to x -axis, then $y_1 = y_2$
and, $PQ = \text{Absolute value of the difference of } x\text{-coordinates of } P \text{ and } Q = |x_2 - x_1|$.
- Let $P(x, 0)$ be a point on x -axis which is equidistant from the points $A(7, 6)$ and $B(3, 4)$. Then,
 $PA = PB$
 $\Rightarrow PA^2 = PB^2$

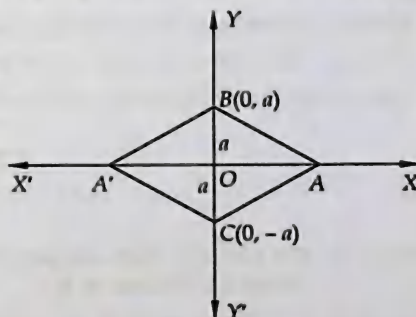


Fig. 22.10

$$\Rightarrow (x-7)^2 + (0-6)^2 = (x-3)^2 + (0-4)^2 \Rightarrow -14x + 85 = -6x + 25 \Rightarrow 8x = 60 \Rightarrow x = \frac{15}{2}$$

Hence, $P\left(\frac{15}{2}, 0\right)$ is the required point.

22.6 LOCUS AND EQUATION TO A LOCUS

LOCUS The curve described by a point which moves under given condition or conditions is called its locus.

For example, suppose C is a point in the plane of the paper and P is a variable point in the plane of the paper such that its distance from C is always equal to a (say). It is clear that all the positions of the moving point P lie on the circumference of a circle whose centre is C and whose radius is a . The circumference of this circle is therefore the "Locus" of point P when it moves under the condition that its distance from the point C is always equal to constant a .

Let A and B be two fixed points in the plane of the paper and P be a variable point in the plane of the paper which moves in such a way that its distances from A and B are always equal. Obviously all the positions of the moving point P lie on the perpendicular bisector of AB . Thus, the "locus" of P is the perpendicular bisector of AB when it moves under the condition that its distances from A and B are always equal.

EQUATION TO THE LOCUS OF A POINT The equation to the locus of a point is the relation which is satisfied by the coordinates of every point on the locus of the point.

In order to find the locus of a point, we may use the following algorithm.

ALGORITHM

STEP I Assume the coordinates of the point say, (h, k) whose locus is to be found.

STEP II Write the given condition in mathematical form involving h, k .

STEP III Eliminate the variable (s) , if any.

STEP IV Replace h by x and k by y in the result obtained in step III.

The equation so obtained is the locus of the point which moves under some stated condition (s) .

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I ON FINDING THE LOCUS OF A POINT WHEN GIVEN GEOMETRICAL CONDITIONS DO NOT INVOLVE A VARIABLE

EXAMPLE 1 The sum of the squares of the distances of a moving point from two fixed points $(a, 0)$ and $(-a, 0)$ is equal to a constant quantity $2c^2$. Find the equation to its locus.

SOLUTION Let $P(h, k)$ be any position of the moving point and let $A(a, 0)$, $B(-a, 0)$ be the given points. Then,

$$PA^2 + PB^2 = 2c^2 \quad \text{[Given]}$$

$$\Rightarrow (h-a)^2 + (k-0)^2 + (h+a)^2 + (k-0)^2 = 2c^2$$

$$\Rightarrow h^2 - 2ah + a^2 + k^2 + h^2 + 2ah + a^2 + k^2 = 2c^2$$

$$\Rightarrow 2h^2 + 2k^2 + 2a^2 = 2c^2$$

$$\Rightarrow h^2 + k^2 = c^2 - a^2$$

Hence, locus of (h, k) is $x^2 + y^2 = c^2 - a^2$.

EXAMPLE 2 Find the equation to the locus of a point equidistant from the points $A(1, 3)$ and $B(-2, 1)$.

SOLUTION Let $P(h, k)$ be any point on the locus. Then,

$$PA = PB \quad \text{[Given]}$$

$$\Rightarrow PA^2 = PB^2 \Rightarrow (h-1)^2 + (k-3)^2 = (h+2)^2 + (k-1)^2 \Rightarrow 6h + 4k = 5$$

Hence, locus of (h, k) is $6x + 4y = 5$.

EXAMPLE 3 Find the locus of a point such that the sum of its distances from the points $(0, 2)$ and $(0, -2)$ is 6.

SOLUTION Let $P(h, k)$ be any point on the locus and let $A(0, 2)$ and $B(0, -2)$ be the given points. By the given condition

$$PA + PB = 6$$

$$\Rightarrow \sqrt{(h-0)^2 + (k-2)^2} + \sqrt{(h-0)^2 + (k+2)^2} = 6$$

$$\Rightarrow \sqrt{h^2 + (k-2)^2} = 6 - \sqrt{h^2 + (k+2)^2}$$

$$\Rightarrow h^2 + (k-2)^2 = 36 - 12\sqrt{h^2 + (k+2)^2} + h^2 + (k+2)^2$$

$$\Rightarrow -8k - 36 = -12\sqrt{h^2 + (k+2)^2}$$

$$\Rightarrow (2k+9) = 3\sqrt{h^2 + (k+2)^2}$$

$$\Rightarrow (2k+9)^2 = 9(h^2 + (k+2)^2)$$

$$\Rightarrow 4k^2 + 36k + 81 = 9h^2 + 9k^2 + 36k + 36$$

$$\Rightarrow 9h^2 + 5k^2 = 45$$

Hence, locus of (h, k) is $9x^2 + 5y^2 = 45$.

EXAMPLE 4 Find the locus of a point, so that the join of $(-5, 1)$ and $(3, 2)$ subtends a right angle at the moving point.

SOLUTION Let $P(h, k)$ be a moving point and let $A(-5, 1)$ and $B(3, 2)$ be given points. By the given condition

$$\angle APB = 90^\circ$$

$\therefore \Delta APB$ is a right angle triangle

$$\Rightarrow AB^2 = AP^2 + PB^2$$

$$\Rightarrow (3+5)^2 + (2-1)^2 = (h+5)^2 + (k-1)^2 + (h-3)^2 + (k-2)^2$$

$$\Rightarrow 65 = 2(h^2 + k^2 + 2h - 3k) + 39$$

$$\Rightarrow h^2 + k^2 + 2h - 3k - 13 = 0$$

Hence, locus of (h, k) is $x^2 + y^2 + 2x - 3y - 13 = 0$.

EXAMPLE 5 A point moves so that the sum of its distances from $(ae, 0)$ and $(-ae, 0)$ is $2a$, prove that the equation to its locus is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b^2 = a^2(1 - e^2)$.

SOLUTION Let $P(h, k)$ be the moving point such that the sum of its distances from $A(ae, 0)$ and $B(-ae, 0)$ is $2a$. Then,

$$PA + PB = 2a$$

$$\Rightarrow \sqrt{(h-ae)^2 + (k-0)^2} + \sqrt{(h+ae)^2 + (k-0)^2} = 2a$$

$$\Rightarrow \sqrt{(h-ae)^2 + k^2} = 2a - \sqrt{(h+ae)^2 + k^2}$$

$$\Rightarrow (h-ae)^2 + k^2 = 4a^2 + (h+ae)^2 + k^2 - 4a\sqrt{(h+ae)^2 + k^2} \quad [\text{Squaring both sides}]$$

$$\Rightarrow -4aeh - 4a^2 = -4a\sqrt{(h+ae)^2 + k^2}$$

$$\begin{aligned}
 \Rightarrow (eh + a) &= \sqrt{(h + ae)^2 + k^2} \\
 \Rightarrow (eh + a)^2 &= (h + ae)^2 + k^2 \\
 \Rightarrow e^2 h^2 + 2ae h + a^2 &= h^2 + a^2 e^2 + 2ae h + k^2 \\
 \Rightarrow h^2 (1 - e^2) + k^2 &= a^2 (1 - e^2) \\
 \Rightarrow \frac{h^2}{a^2} + \frac{k^2}{a^2 (1 - e^2)} &= 1
 \end{aligned}$$

Hence, locus of (h, k) is $\frac{x^2}{a^2} + \frac{y^2}{a^2 (1 - e^2)} = 1$ or, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b^2 = a^2 (1 - e^2)$

EXAMPLE 6 Find the equation to the locus of a point which moves so that the sum of its distances from $(3, 0)$ and $(-3, 0)$ is less than 9.

SOLUTION Let $P(h, k)$ be the moving point such that the sum of its distances from $A(3, 0)$ and $B(-3, 0)$ is less than 9. Then,

$$PA + PB < 9$$

$$\begin{aligned}
 \Rightarrow \sqrt{(h-3)^2 + (k-0)^2} + \sqrt{(h+3)^2 + (k-0)^2} &< 9 \\
 \Rightarrow \sqrt{(h-3)^2 + k^2} &< 9 - \sqrt{(h+3)^2 + k^2} \\
 \Rightarrow (h-3)^2 + k^2 &< [9 - \sqrt{(h+3)^2 + k^2}]^2 \quad [\because x < y \Rightarrow x^2 < y^2 \text{ for } x, y > 0] \\
 \Rightarrow (h-3)^2 + k^2 &< 81 + (h+3)^2 + k^2 - 18\sqrt{(h+3)^2 + k^2} \\
 \Rightarrow -12h - 81 &< -18\sqrt{(h+3)^2 + k^2} \\
 \Rightarrow 4h + 27 &> 6\sqrt{(h+3)^2 + k^2} \\
 \Rightarrow (4h + 27)^2 &> 36[(h+3)^2 + k^2] \\
 \Rightarrow 20h^2 + 36k^2 &< 405
 \end{aligned}$$

Hence, the locus of (h, k) is $20x^2 + 36y^2 < 405$.

LEVEL-2

Type II ON FINDING THE LOCUS OF A POINT WHEN GIVEN GEOMETRICAL CONDITIONS INVOLVE SOME VARIABLE(S)

EXAMPLE 7 Find the locus of the point of intersection of the lines $x \cos \alpha + y \sin \alpha = a$ and $x \sin \alpha - y \cos \alpha = b$, where α is a variable.

SOLUTION Let $P(h, k)$ be the point of intersection of the given lines. Then,

$$h \cos \alpha + k \sin \alpha = a \quad \dots(i)$$

$$\text{and, } h \sin \alpha - k \cos \alpha = b \quad \dots(ii)$$

Here α is a variable. So, we have to eliminate α . Squaring and adding (i) and (ii), we get

$$(h \cos \alpha + k \sin \alpha)^2 + (h \sin \alpha - k \cos \alpha)^2 = a^2 + b^2 \Rightarrow h^2 + k^2 = a^2 + b^2$$

Hence, the locus of (h, k) is $x^2 + y^2 = a^2 + b^2$.

EXAMPLE 8 A rod of length l slides with its ends on two perpendicular lines. Find the locus of its mid-point.

SOLUTION Let the two perpendicular lines be the coordinate axes. Let AB be a rod of length l . Let the coordinates of A and B be $(a, 0)$ and $(0, b)$ respectively. As the rod slides the values of a and b change. So a and b are two variables. Let $P(h, k)$ be the mid-point of the rod AB in one of the infinite positions it attains. Then,

$$h = \frac{a+0}{2} \text{ and } k = \frac{0+b}{2}$$

$$\Rightarrow h = \frac{a}{2} \text{ and } k = \frac{b}{2} \quad \dots(i)$$

From $\triangle OAB$, we have

$$AB^2 = OA^2 + OB^2$$

$$\Rightarrow a^2 + b^2 = l^2$$

$$\Rightarrow (2h)^2 + (2k)^2 = l^2 \quad [\text{Using (i)}]$$

$$\Rightarrow 4h^2 + 4k^2 = l^2$$

Hence, the locus of (h, k) is $4x^2 + 4y^2 = l^2$.

EXAMPLE 9 AB is a variable line sliding between the coordinate axes in such a way that A lies on x -axis and B lies on y -axis. If P is a variable point on AB such that $PA = b$, $PB = a$ and $AB = a + b$, find the equation of the locus of P .

SOLUTION Let $P(h, k)$ be a variable point on AB such that $\angle OAB = \theta$, where θ is a variable. From triangles ALP and PMB , we have

$$\sin \theta = \frac{k}{b} \quad \dots(i) \quad \cos \theta = \frac{h}{a} \quad \dots(ii)$$

Here, θ is a variable. So, we have to eliminate θ .

Squaring (i) and (ii) and adding, we get: $\frac{k^2}{b^2} + \frac{h^2}{a^2} = 1$

Hence, the locus of (h, k) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

EXAMPLE 10 If O is the origin and Q is a variable point on $x^2 = 4y$. Find the locus of the mid-point of OQ .

SOLUTION Let the coordinates of Q be (a, b) and let $P(h, k)$ be the mid-point of OQ . Then,

$$h = \frac{a+0}{2} = \frac{a}{2} \text{ and } k = \frac{0+b}{2} = \frac{b}{2} \Rightarrow a = 2h \text{ and } b = 2k \quad \dots(i)$$

Here, a and b are two variables which are to be eliminated. Since, (a, b) lies on $x^2 = 4y$.

$$\therefore a^2 = 4b$$

$$\Rightarrow (2h)^2 = 4(2k)$$

$$\Rightarrow h^2 = 2k$$

Hence, the locus of (h, k) is $x^2 = 2y$.

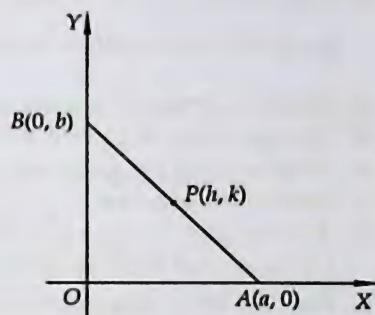


Fig. 22.11

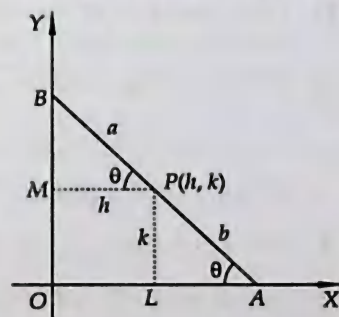


Fig. 22.12

[Using (i)]

EXERCISE 22.2

LEVEL-1

- Find the locus of a point equidistant from the point (2, 4) and the y -axis.
- Find the equation of the locus of a point which moves such that the ratio of its distances from (2, 0) and (1, 3) is 5 : 4.
- A point moves as so that the difference of its distances from $(ae, 0)$ and $(-ae, 0)$ is $2a$, prove that the equation to its locus is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $b^2 = a^2(e^2 - 1)$.
- Find the locus of a point such that the sum of its distances from (0, 2) and (0, -2) is 6.
- Find the locus of a point which is equidistant from (1, 3) and x -axis.
- Find the locus of a point which moves such that its distance from the origin is three times its distance from x -axis.
- $A(5, 3)$, $B(3, -2)$ are two fixed points; find the equation to the locus of a point P which moves so that the area of the triangle PAB is 9 units.
- Find the locus of a point such that the line segments having end points (2, 0) and (-2, 0) subtend a right angle at that point.
- If $A(-1, 1)$ and $B(2, 3)$ are two fixed points, find the locus of a point P so that the area of $\triangle PAB = 8$ sq. units.

LEVEL-2

- A rod of length l slides between the two perpendicular lines. Find the locus of the point on the rod which divides it in the ratio 1 : 2.
- Find the locus of the mid-point of the portion of the line $x \cos \alpha + y \sin \alpha = p$ which is intercepted between the axes.
- If O is the origin and Q is a variable point on $y^2 = x$. Find the locus of the mid-point of OQ .

ANSWERS

- $y^2 - 8y - 4x + 20 = 0$
- $9x^2 + 9y^2 + 14x - 150y + 186 = 0$
- $9x^2 + 5y^2 = 45$
- $x^2 - 2x - 6y + 10 = 0$
- $x^2 = 8y^2$
- $5x - 2y - 1 = 0$ or $5x - 2y - 37 = 0$
- $x^2 + y^2 = 4$
- $2x - 3y - 11 = 0, 2x - 3y + 21 = 0$
- $\frac{x^2}{4} + y^2 = \frac{l^2}{9}$
- $p^2(x^2 + y^2) = 4x^2y^2$
- $2y^2 = x$

HINTS TO SELECTED PROBLEM

- Let $P(h, k)$ be the variable point and let $A(2, 0)$ and $B(-2, 0)$ be the given points. Then, $\angle APB = \pi/2 \Rightarrow AB^2 = PA^2 + PB^2$.

22.7 SHIFTING OF ORIGIN

Let O be the origin and let $x'Ox$ and $y'Oy$ be the axis of x and y respectively. Let O' and P be two points in the plane having coordinates (h, k) and (x, y) respectively referred to $X'OX$ and $Y'OY$ as the coordinate axes. Let the origin be transferred to O' and let $X'O'X$ and $Y'O'Y$ be new rectangular axes. Let the coordinates of P referred to new axes as the coordinate axes be (X, Y) . Then,

$$O'N = X, PN = Y, OM = x, PM = y, OL = h \text{ and, } O'L = k.$$

Now, $x = OM = OL + LM = OL + O'N = h + X$

and, $y = PM = PN + NM = PN + O'L = Y + k$

$\therefore x = X + h$ and $y = Y + k$.

Thus, if (x, y) are coordinates of a point referred to old axes and (X, Y) are the coordinates of the same point referred to new axes, then

$$x = X + h \text{ and } y = Y + k$$

i.e., (Old x -coordinate) = (New x -coordinate) + h

and, (Old y -coordinate) = (New y -coordinate) + k .

If therefore the origin is shifted at a point (h, k) we must substitute $X + h$ and $Y + k$ for x and y respectively.

The transformation formula from new axes to old axes is: $X = x - h, Y = y - k$

The coordinates of the old origin referred to the new axes are $(-h, -k)$.

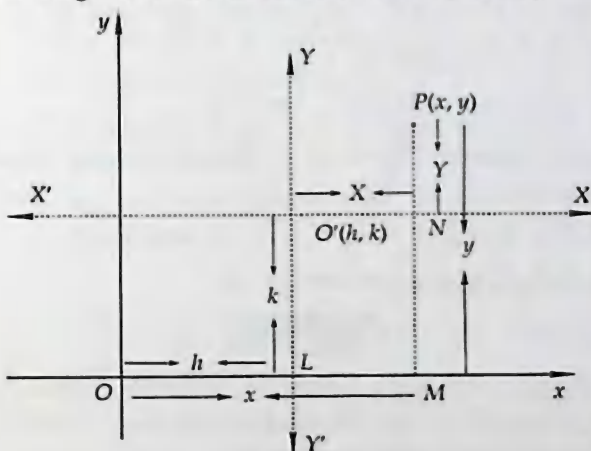


Fig. 22.13

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 If the axes are shifted to the point $(1, -2)$ without rotation, what do the following equations become?

(i) $2x^2 + y^2 - 4x + 4y = 0$

(ii) $y^2 - 4x + 4y + 8 = 0$

SOLUTION (i) Substituting $x = X + 1, y = Y + (-2) = Y - 2$ in the equation $2x^2 + y^2 - 4x + 4y = 0$, we get

$$2(X+1)^2 + (Y-2)^2 - 4(X+1) + 4(Y-2) = 0 \Rightarrow 2X^2 + Y^2 = 6.$$

(ii) Substituting $x = X + 1, y = Y - 2$ in the equation $y^2 - 4x + 4y + 8 = 0$, we get

$$(Y-2)^2 - 4(X+1) + 4(Y-2) + 8 = 0 \Rightarrow Y^2 = 4X$$

EXAMPLE 2 At what point the origin be shifted, if the coordinates of a point $(4, 5)$ become $(-3, 9)$?

SOLUTION Let (h, k) be the point to which the origin is shifted. Then,

$$x = 4, y = 5, X = -3, Y = 9$$

$$\therefore x = X + h \text{ and } y = Y + k \Rightarrow 4 = -3 + h \text{ and } 5 = 9 + k \Rightarrow h = 7 \text{ and } k = -4$$

Hence, the origin must be shifted to $(7, -4)$

EXAMPLE 3 Shift the origin to a suitable point so that the equation $y^2 + 4y + 8x - 2 = 0$ will not contain term in y and the constant term.

SOLUTION Let the origin be shifted to (h, k) . Then, $x = X + h$ and $y = Y + k$.

Substituting $x = X + h$, $y = Y + k$ in the equation $y^2 + 4y + 8x - 2 = 0$, we get

$$(Y + k)^2 + 4(Y + k) + 8(X + h) - 2 = 0$$

$$\Rightarrow Y^2 + (4 + 2k)Y + 8X + (k^2 + 4k + 8h - 2) = 0$$

For this equation to be free from the term containing Y and the constant term, we must have

$$4 + 2k = 0 \text{ and } k^2 + 4k + 8h - 2 = 0 \Rightarrow k = -2 \text{ and } h = \frac{3}{4}.$$

Hence, the origin is shifted at the point $(3/4, -2)$.

EXAMPLE 4 Find the point to which the origin should be shifted so that the equation $y^2 - 6y - 4x + 13 = 0$ is transformed to the form $y^2 + Ax = 0$.

SOLUTION Let the origin be shifted to the point (h, k) . Then, $x = X + h$ and $y = Y + k$.

Substituting $x = X + h$ and $y = Y + k$ in the equation $y^2 - 6y - 4x + 13 = 0$, we get

$$(Y + k)^2 - 6(Y + k) - 4(X + h) + 13 = 0$$

$$\Rightarrow Y^2 + (2k - 6)Y - 4X + (k^2 - 6k + 13 - 4h) = 0$$

This equation should be of the form $Y^2 + AX = 0$. This means that it should not contain term containing Y and constant term.

$$\therefore 2k - 6 = 0 \text{ and } k^2 - 4h + 13 - 6k = 0 \Rightarrow k = 3 \text{ and } h = 1$$

Hence, the coordinates of the required point are $(1, 3)$.

LEVEL-2

EXAMPLE 5 Prove that the area of a triangle is invariant under the translation of the axes.

SOLUTION Let ABC be a triangle having the coordinates of its vertices as $A(x_1, y_1)$, $B(x_2, y_1)$ and $C(x_3, y_3)$. Then, area of triangle ABC is given by

$$\Delta = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)] \quad \dots(i)$$

Let the origin be shifted at (h, k) . Then, the new coordinates of the vertices are

$$A(x_1 + h, y_1 + k), B(x_2 + h, y_2 + k) \text{ and } C(x_3 + h, y_3 + k).$$

Therefore, the area of the triangle in the new-coordinate system is given by

$$\Delta_1 = \frac{1}{2} \left[(x_1 + h) \{(y_2 + k) - (y_3 + k)\} + (x_2 + h) \{(y_3 + k) - (y_1 + k)\} \right. \\ \left. + (x_3 + h) \{(y_1 + k) - (y_2 + k)\} \right]$$

$$\Rightarrow \Delta_1 = \frac{1}{2} \left[(x_1 + h)(y_2 - y_3) + (x_2 + h)(y_3 - y_1) + (x_3 + h)(y_1 - y_2) \right]$$

$$\Rightarrow \Delta_1 = \frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) + h(y_2 - y_3 + y_3 - y_1 + y_1 - y_2) \right]$$

$$\Rightarrow \Delta_1 = \frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right] \quad \dots(ii)$$

From (i) and (ii), we get $\Delta = \Delta_1$.

Hence, the area of a triangle is invariant under the translation of the axes.

EXERCISE 22.3

LEVEL-1

- What does the equation $(x-a)^2 + (y-b)^2 = r^2$ become when the axes are transferred to parallel axes through the point $(a-c, b)$?
- What does the equation $(a-b)(x^2 + y^2) - 2abx = 0$ become if the origin is shifted to the point $\left(\frac{ab}{a-b}, 0\right)$ without rotation?
- Find what the following equations become when the origin is shifted to the point $(1, 1)$
 - $x^2 + xy - 3x - y + 2 = 0$
 - $x^2 - y^2 - 2x + 2y = 0$
 - $xy - x - y + 1 = 0$
 - $xy - y^2 - x + y = 0$
- At what point the origin be shifted so that the equation $x^2 + xy - 3x - y + 2 = 0$ does not contain any first degree term and constant term?
- Verify that the area of the triangle with vertices $(2, 3)$, $(5, 7)$ and $(-3, -1)$ remains invariant under the translation of axes when the origin is shifted to the point $(-1, 3)$.
- Find, what the following equations become when the origin is shifted to the point $(1, 1)$
 - $x^2 + xy - 3y^2 - y + 2 = 0$
 - $xy - y^2 - x + y = 0$
 - $xy - x - y + 1 = 0$
 - $x^2 - y^2 - 2x + 2y = 0$
- Find the point to which the origin should be shifted after a translation of axes so that the following equations will have no first degree terms:
 - $y^2 + x^2 - 4x - 8y + 3 = 0$
 - $x^2 + y^2 - 5x + 2y - 5 = 0$
 - $x^2 - 12x + 4 = 0$
- Verify that the area of the triangle with vertices $(4, 6)$, $(7, 10)$ and $(1, -2)$ remains invariant under the translation of axes when the origin is shifted to the point $(-2, 1)$. [NCERT]

ANSWERS

- $X^2 + Y^2 - 2CX = r^2 - c^2$
- $(a-b)^2 (X^2 + Y^2) = a^2 b^2$
- $x^2 + xy = 0$
 - $x^2 - y^2 = 0$
 - $xy = 0$
 - $xy - y^2 = 0$
- $(1, 1)$
- $x^2 - 3y^2 + xy + 3x - 6y = 0$
 - $xy - y^2 = 0$
- $xy = 0$
 - $x^2 - y^2 = 0$
 - $(2, 4)$
 - $(5/2, -1)$
 - $(6, k), k \in \mathbb{R}$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

- The vertices of a triangle are $O(0, 0)$, $A(a, 0)$ and $B(0, b)$. Write the coordinates of its circumcentre.
- In Q.No. 1, write the distance between the circumcentre and orthocentre of $\triangle OAB$.
- Write the coordinates of the orthocentre of the triangle formed by points $(8, 0)$, $(4, 6)$ and $(0, 0)$.
- Three vertices of a parallelogram, taken in order, are $(-1, -6)$, $(2, -5)$ and $(7, 2)$. Write the coordinates of its fourth vertex.

5. If the points $(a, 0)$, $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ are collinear, write the value of $t_1 t_2$.
6. If the coordinates of the sides AB and AC of a $\triangle ABC$ are $(3, 5)$ and $(-3, -3)$ respectively, then write the length of side BC .
7. Write the coordinates of the circumcentre of a triangle whose centroid and orthocentre are at $(3, 3)$ and $(-3, 5)$ respectively.
8. Write the coordinates of the incentre of the triangle having its vertices at $(0, 0)$, $(5, 0)$ and $(0, 12)$.
9. If the points $(1, -1)$, $(2, -1)$ and $(4, -3)$ are the mid-points of the sides of a triangle, then write the coordinates of its centroid.
10. Write the area of the triangle having vertices at $(a, b+c)$, $(b, c+a)$, $(c, a+b)$.

ANSWERS

1. $\left(\frac{a}{2}, \frac{b}{2}\right)$
2. $\frac{1}{2}\sqrt{a^2+b^2}$
3. $\left(\frac{8}{3}, 4\right)$
4. $(4, 1)$
5. -1
6. 20 units
7. $(6, 2)$
8. $(2, 2)$
9. $\left(\frac{7}{3}, \frac{-5}{3}\right)$
10. 0

SUMMARY

1. The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
 i.e. $PQ = \sqrt{(\text{Difference of abscissae})^2 + (\text{Difference of ordinates})^2}$
2. The distance of a point $P(x, y)$ from the origin $O(0, 0)$ is given by $OP = \sqrt{x^2 + y^2}$.
3. The area of the triangle, the coordinates of whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) is the absolute value of

$$\frac{1}{2} \left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right] \text{ or, } \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$
4. If the points (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are collinear, then

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$
5. The coordinates of the point dividing the line segment joining (x_1, y_1) and (x_2, y_2) in the ratio $m:n$ are

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right), \text{ internally; } \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right), \text{ externally}$$
6. The coordinates of the mid-point of the line segment joining (x_1, y_1) and (x_2, y_2) are

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$
7. The coordinates of the centroid of the triangle whose vertices are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$$

THE STRAIGHT LINES

23.1 DEFINITION OF A STRAIGHT LINE

A straight line is a curve such that every point on the line segment joining any two points on it lies on it.

THEOREM Every first degree equation in x, y represents a straight line.

[NCERT EXEMPLAR]

PROOF Let $ax + by + c = 0$ be a first degree equation in x, y where a, b, c are constants. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be any two points on the curve represented by $ax + by + c = 0$. Then,

$$ax_1 + by_1 + c = 0 \text{ and } ax_2 + by_2 + c = 0 \quad \dots(i)$$

Let R be any point on the line segment joining P and Q . Suppose R divides PQ in the ratio $\lambda : 1$.

Then, the coordinates of R are $\left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1}\right)$. In order to prove that the curve

represented by $ax + by + c = 0$ is a straight line, it is sufficient to show that R lies on it for all values of λ .

Now,

$$\begin{aligned} a\left(\frac{\lambda x_2 + x_1}{\lambda + 1}\right) + b\left(\frac{\lambda y_2 + y_1}{\lambda + 1}\right) + c &= \frac{\lambda(ax_2 + by_2 + c) + (ax_1 + by_1 + c)}{\lambda + 1} \\ &= \lambda \cdot 0 + 0 = 0 \end{aligned}$$

[Using (i)]

$\therefore R\left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1}\right)$ lies on the curve represented by $ax + by + c = 0$.

Thus, every point on the line segment joining P and Q lies on $ax + by + c = 0$.

Hence, $ax + by + c = 0$ represents a straight line.

Q.E.D.

NOTE When we say that a first degree equation in x, y i.e., $ax + by + c = 0$ represents a line, it means that all points (x, y) satisfying $ax + by + c = 0$ lie along a line. Thus, a line is also defined as the locus of a point satisfying the condition $ax + by + c = 0$ where a, b, c are constants.

It follows from the above discussion that $ax + by + c = 0$ is the general equation of a line.

It should be noted that there are only two unknowns in the equation of a straight line because equation of every straight line can be put in the form $ax + by + 1 = 0$ where a, b are two unknowns. Note that x, y are not unknowns. In fact these are the coordinates of any point on the line and are known as the current coordinates. Thus, to determine a line we will need two conditions to determine the two unknowns. In the further discussion on straight line you will find that whenever it will be asked to find a straight line there will always be two conditions connecting the two unknowns.

23.2 SLOPE (GRADIENT) OF A LINE

The trigonometrical tangent of the angle that a line makes with the positive direction of the x -axis in anticlockwise sense is called the slope or gradient of the line.

The slope of a line is generally denoted by m . Thus, $m = \tan \theta$.

Since a line parallel to x -axis makes an angle of 0° with x -axis, therefore its slope is $\tan 0^\circ = 0$. A line parallel to y -axis i.e., perpendicular to x -axis makes an angle of 90° with x -axis, so its slope is

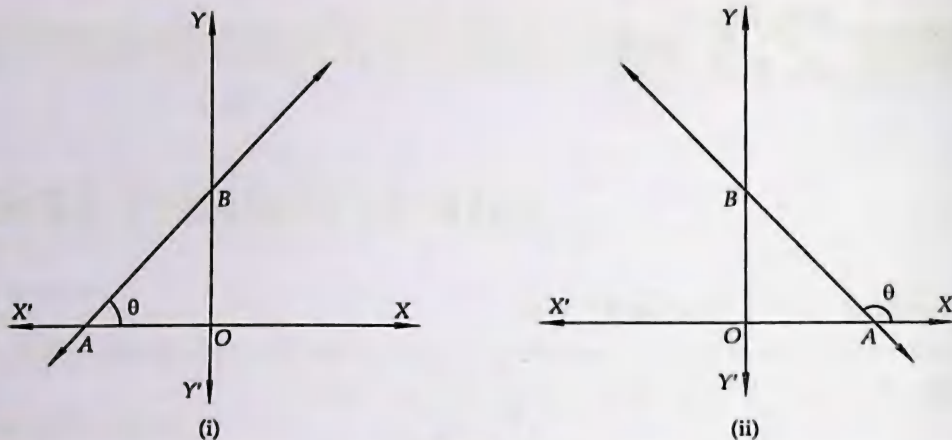


Fig. 23.1

$\tan \pi/2 = \infty$. Also, the slope of a line equally inclined with axes is 1 or -1 as it makes 45° or 135° angle with x -axis.

REMARK The angle of inclination of a line with the positive direction of x -axis in anticlockwise sense always lies between 0° and 180° .

ILLUSTRATION 1 Find the slope of a line whose inclination to the positive direction of x -axis in anticlockwise sense is (i) 60° (ii) 0° (iii) 150° (iv) 120° . [NCERT]

SOLUTION (i) Slope $= \tan 60^\circ = \sqrt{3}$.

(ii) Slope $= \tan 0^\circ = 0$.

(iii) Slope $= \tan 150^\circ = -\cot 60^\circ = -\frac{1}{\sqrt{3}}$.

(iv) Slope $= \tan 120^\circ = -\cot 30^\circ = -\sqrt{3}$.

ILLUSTRATION 2 What can be said regarding a line if its slope is (i) positive (ii) zero (iii) negative?

SOLUTION Let θ be the angle of inclination of the given line with the positive direction of x -axis in anticlockwise sense. Then, its slope is given by $m = \tan \theta$.

(i) If the slope of the line is positive, then

$$m = \tan \theta > 0 \Rightarrow \theta \text{ lies between } 0^\circ \text{ and } 90^\circ \Rightarrow \theta \text{ is an acute angle.}$$

Thus, a line of positive slope makes an acute angle with the positive direction of x -axis.

(ii) If the slope of the line is zero, then

$$m = \tan \theta = 0 \Rightarrow \theta = 0^\circ \Rightarrow \text{either the line is } x\text{-axis or it is parallel to } x\text{-axis.}$$

Thus, a line of zero slope is parallel or coincident to x -axis.

(iii) If the slope of the line is negative, then

$$m = \tan \theta < 0 \Rightarrow \theta \text{ lies between } 90^\circ \text{ and } 180^\circ \Rightarrow \theta \text{ is an obtuse angle.}$$

Thus, a line of negative slope makes an obtuse angle with the positive direction of x -axis in anticlockwise direction.

23.2.1 SLOPE OF A LINE IN TERMS OF COORDINATES OF ANY TWO POINTS ON IT

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points on a line making an angle θ with the positive direction of x -axis. Draw PL , QM perpendiculars on x -axis and $PN \perp$ on QM . Then,

$$PN = LM = OM - OL = x_2 - x_1 \text{ and, } QN = QM - NM = QM - PL = y_2 - y_1$$

In ΔPQN , we have

$$\tan \theta = \frac{QN}{PN} = \frac{y_2 - y_1}{x_2 - x_1}$$

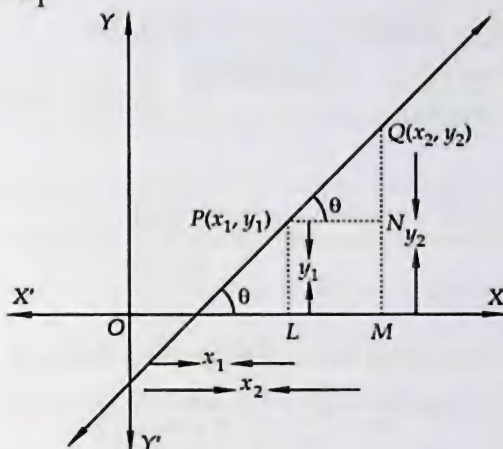


Fig. 23.2

Thus, if (x_1, y_1) and (x_2, y_2) are coordinates of any two points on a line, then its slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Difference of ordinates}}{\text{Difference of abscissae}}$$

ILLUSTRATION 3 Find the slope of a line which passes through points $(3, 2)$ and $(-1, 5)$.

[NCERT]

SOLUTION We know that the slope of a line passing through two points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$. Here, the line passes through $(3, 2)$ and $(-1, 5)$.

So, its slope is given by $m = \frac{5 - 2}{-1 - 3} = -\frac{3}{4}$.

23.3 ANGLE BETWEEN TWO LINES

THEOREM The angle θ between the lines having slopes m_1 and m_2 is given by $\tan \theta = \pm \frac{m_2 - m_1}{1 + m_1 m_2}$.

PROOF Let m_1 and m_2 be the slopes of two given lines AB and CD which intersect at a point P and make angles θ_1 and θ_2 respectively with the positive direction of x -axis. Then, $m_1 = \tan \theta_1$ and $m_2 = \tan \theta_2$.

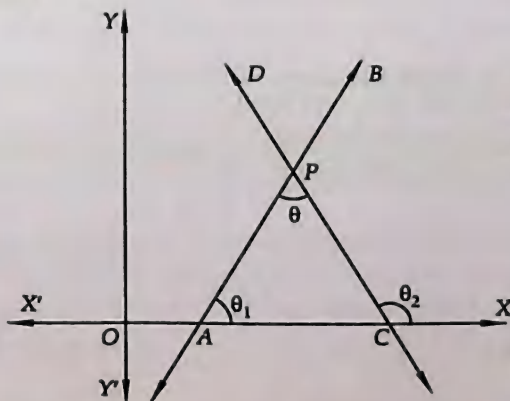


Fig. 23.3

Let $\angle APC = \theta$ be the angle between the given lines. Then,

$$\begin{aligned} \theta_2 &= \theta + \theta_1 && [\text{See Fig. 23.3}] \\ \Rightarrow \theta &= \theta_2 - \theta_1 \\ \Rightarrow \tan \theta &= \tan (\theta_2 - \theta_1) \\ \Rightarrow \tan \theta &= \frac{\tan \theta_2 - \tan \theta_1}{1 + \tan \theta_2 \tan \theta_1} && \left[\text{Using : } \tan (A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} \right] \\ \Rightarrow \tan \theta &= \frac{m_2 - m_1}{1 + m_1 m_2} && \dots(i) \end{aligned}$$

Since $\angle APD = \pi - \theta$ is also the angle between AB and CD . Therefore,

$$\tan \angle APD = \tan (\pi - \theta) = -\tan \theta = -\frac{m_2 - m_1}{1 + m_1 m_2} \quad [\text{Using (i)}] \quad \dots(ii)$$

From (i) and (ii), we find that the angles between two lines of slopes m_1 and m_2 are given by

$$\tan \theta = \pm \frac{m_2 - m_1}{1 + m_1 m_2} \Rightarrow \theta = \tan^{-1} \left(\pm \frac{m_2 - m_1}{1 + m_1 m_2} \right)$$

The acute angle between the lines is given by $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$.

Q.E.D.

ILLUSTRATION If $A(-2, 1)$, $B(2, 3)$ and $C(-2, -4)$ are three points, find the angle between BA and BC .

SOLUTION Let m_1 and m_2 be the slopes of BA and BC respectively. Then,

$$m_1 = \frac{3-1}{2-(-2)} = \frac{2}{4} = \frac{1}{2}, \quad \text{and} \quad m_2 = \frac{-4-3}{-2-2} = \frac{7}{4}$$

Let θ be the acute angle between BA and BC . Then,

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{\frac{7}{4} - \frac{1}{2}}{1 + \frac{7}{4} \times \frac{1}{2}} \right| = \left| \frac{\frac{10}{8}}{\frac{15}{8}} \right| = \frac{2}{3} \Rightarrow \theta = \tan^{-1} \left(\frac{2}{3} \right).$$

CONDITION OF PARALLELISM OF LINES If two lines of slopes m_1 and m_2 are parallel, then the angle θ between them is of 0° .

$$\therefore \tan \theta = \tan 0^\circ = 0$$

$$\Rightarrow \frac{m_2 - m_1}{1 + m_1 m_2} = 0 \quad \left[\text{Using : } \tan \theta = \pm \frac{m_2 - m_1}{1 + m_1 m_2} \right]$$

$$\Rightarrow m_2 = m_1$$

Thus, when two lines are parallel, their slopes are equal.

CONDITION OF PERPENDICULARITY OF TWO LINES If two lines of slopes m_1 and m_2 are perpendicular, then the angle θ between them is of 90° .

From Fig. 23.3, we have

$$\begin{aligned} \theta_2 &= \theta + \theta_1 \\ \Rightarrow \theta_2 &= 90^\circ + \theta_1 && [\because \theta = 90^\circ] \\ \Rightarrow \tan \theta_2 &= \tan (90^\circ + \theta_1) \\ \Rightarrow \tan \theta_2 &= -\cot \theta_1 \end{aligned}$$